Terraform Project

📌 Project Overview

This project uses **Terraform** to provision and manage AWS infrastructure in an automated way. The setup creates a complete environment including EC2 instances, security groups, and supporting AWS resources..

This prject is suitable for **learning Terraform**, **AWS infrastructure automation**, and **DevOps practices**.

In this project implement the following architecture via terraform.

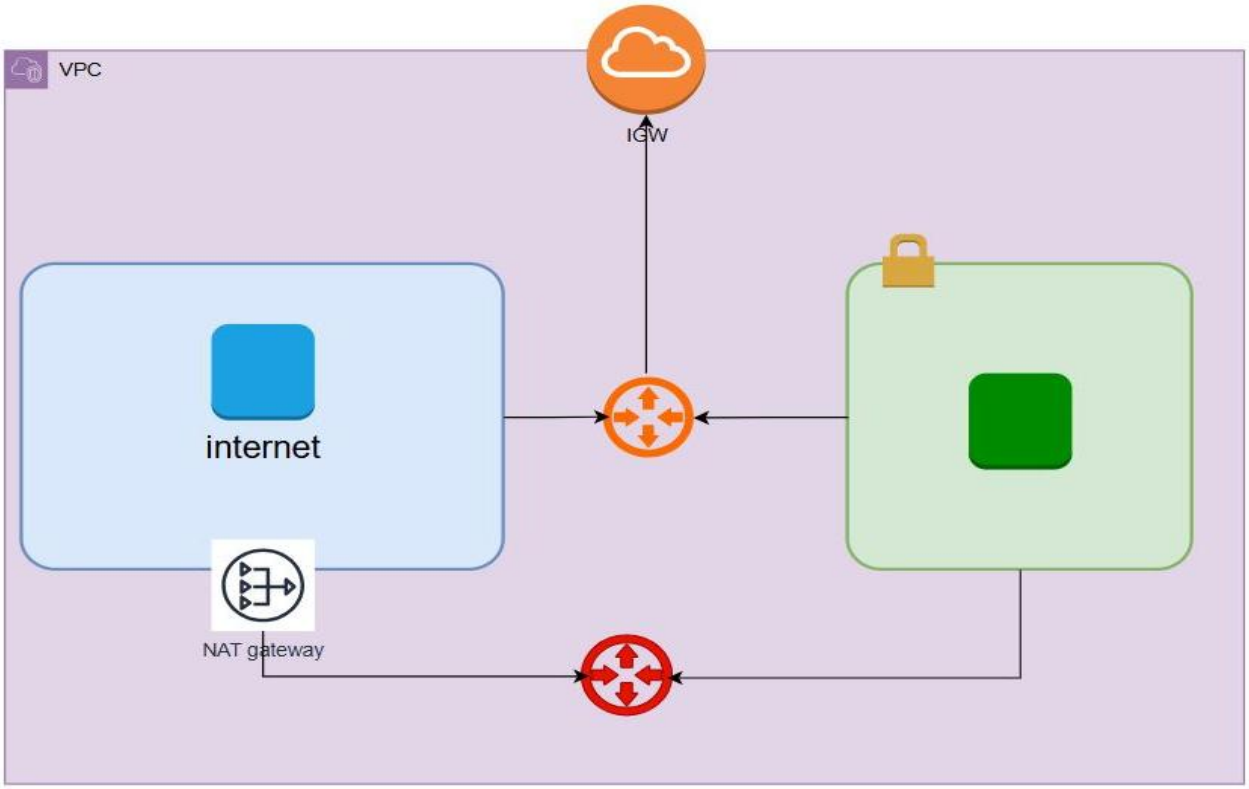


Fig.Architecture Diagram

🛠️ Technologies Used

- **Terraform** (Infrastructure as Code)
- **Amazon Web Services (AWS)**
  - EC2
  - VPC
  - Security Groups
  - Subnet
  - Internet Gateway
  - Route table
  - NAT gateway
- **Mobaxtream (for getting access of instance)**

Prerequisites

Before running this project, make sure you have:

- An **AWS account**
- **Terraform installed** (v1.x recommended)
- **AWS CLI installed and configured**

Project Structure.

```
├── main.tf              # Main Terraform resources definition
├── vars.tf              # Variable declarations
├── terraform.tfvars     # Variable values
├── awsprofile.tf        # AWS provider & profile configuration
└── README.md            # Project documentation
```

Review Project Files

Important files:

- awsprofile.tf → AWS provider & profile configuration
- vars.tf → Input variable definitions
- terraform.tfvars → Variable values (VPC CIDR, instance type, etc.)
- main.tf → Core AWS resources (EC2, ALB, Security Groups)

1. Initialize Terraform

Downloads required providers and initializes backend:

terraform init

2. Validate Terraform Configuration

Checks syntax and configuration correctness:

terraform validate

3. Preview Infrastructure Changes:

Shows what resources will be created:

terraform plan

4. Apply Terraform Configuration

Creates AWS infrastructure:

terraform apply--auto-approve

5. Apply Destroy Configuration

destroy AWS infrastructure:

terraform destroy--auto-approve

Project step-by-step-execution.

1. after terraform apply create the infrastructure in AWS.
2. create 2 instances.
3. get access of public instance via mobaxtream. Then <ping google.com> for check internet.
4. get access of private instance with the help of cmd:

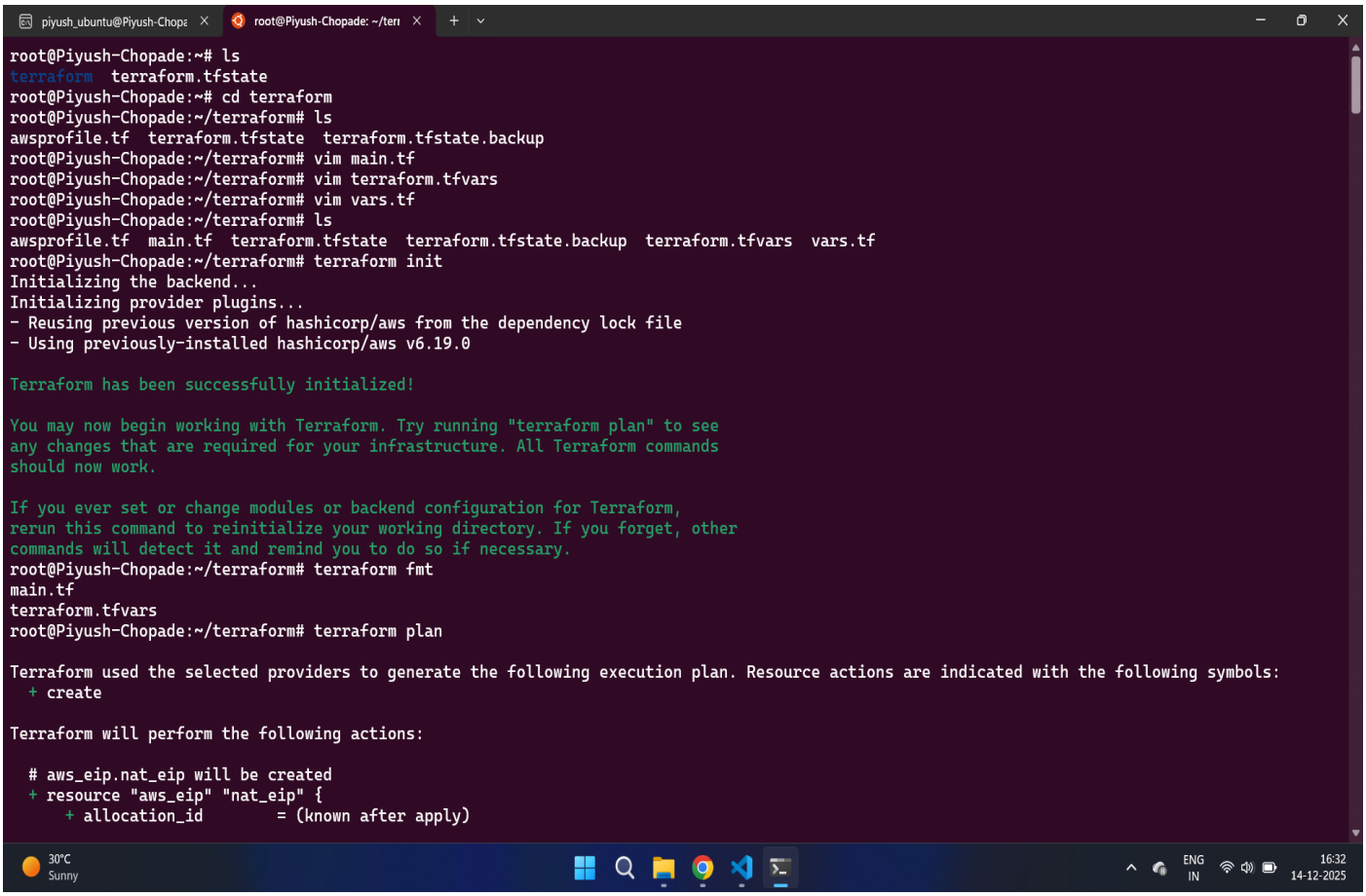
   ssh -i <path-your-key.pem> <machine name>@<PRIVATE_EC2_PRIVATE_IP>

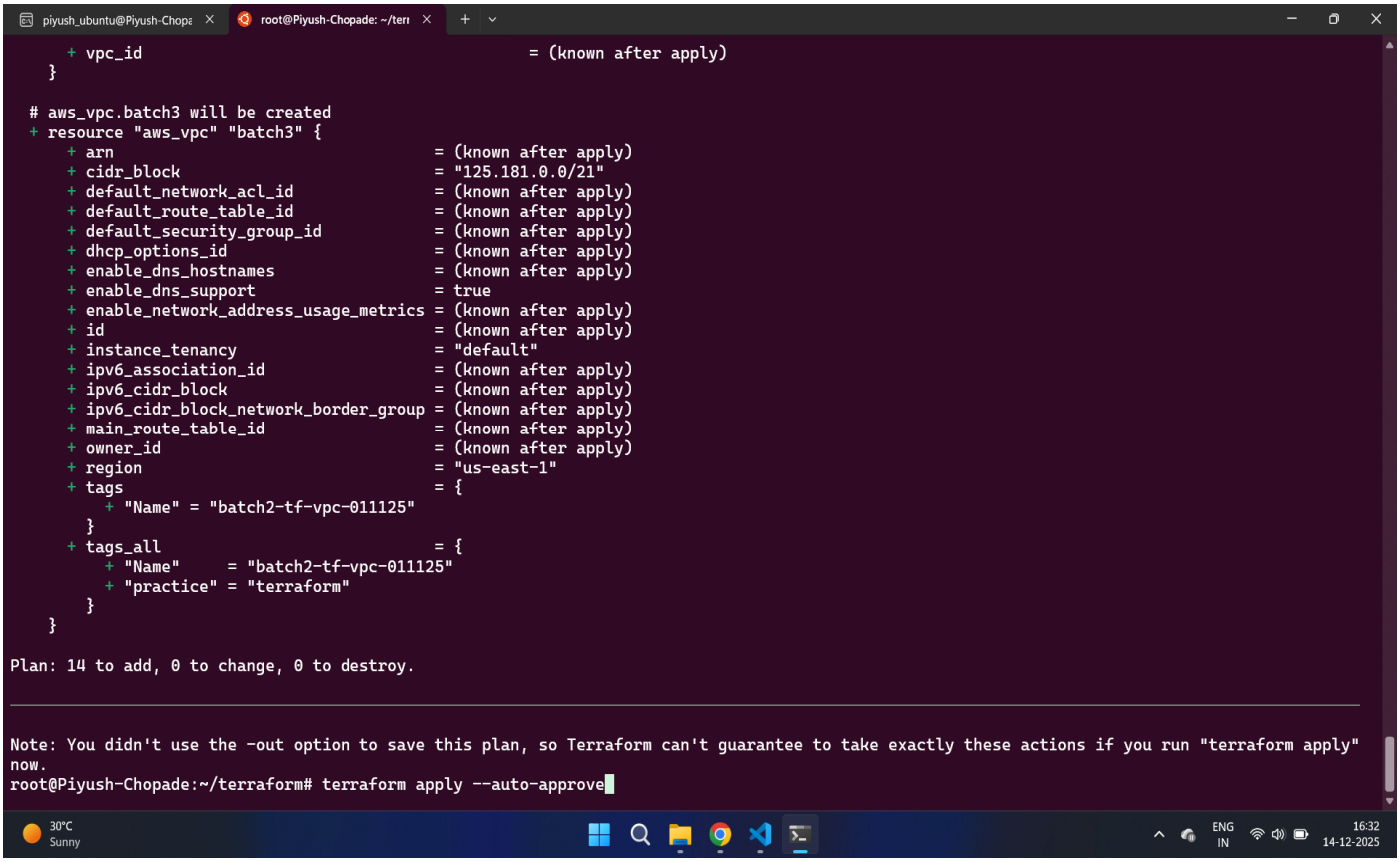
5. give permission to key pair of private instance using cmd:

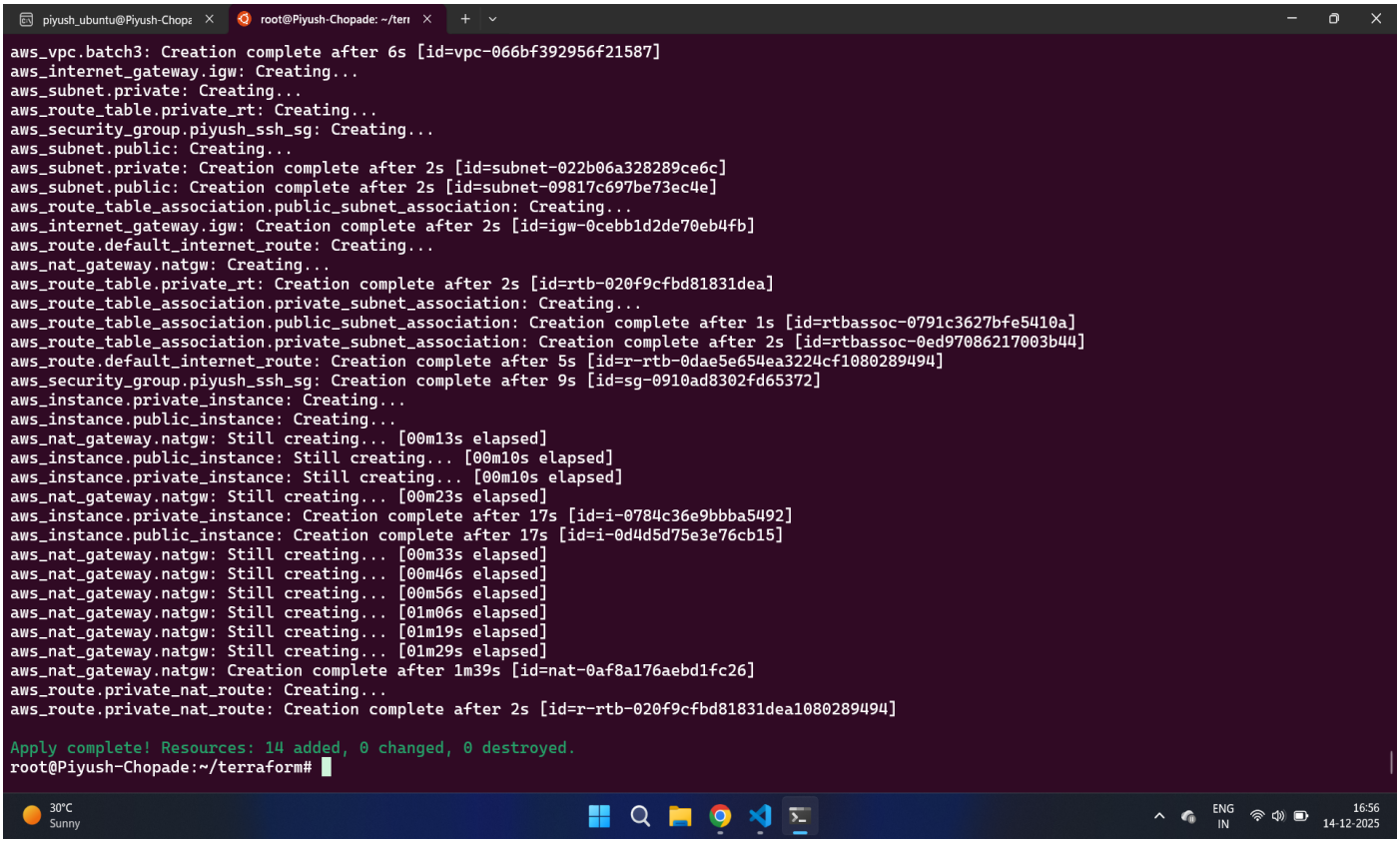
   chmod 400 <path-your-key.pem>

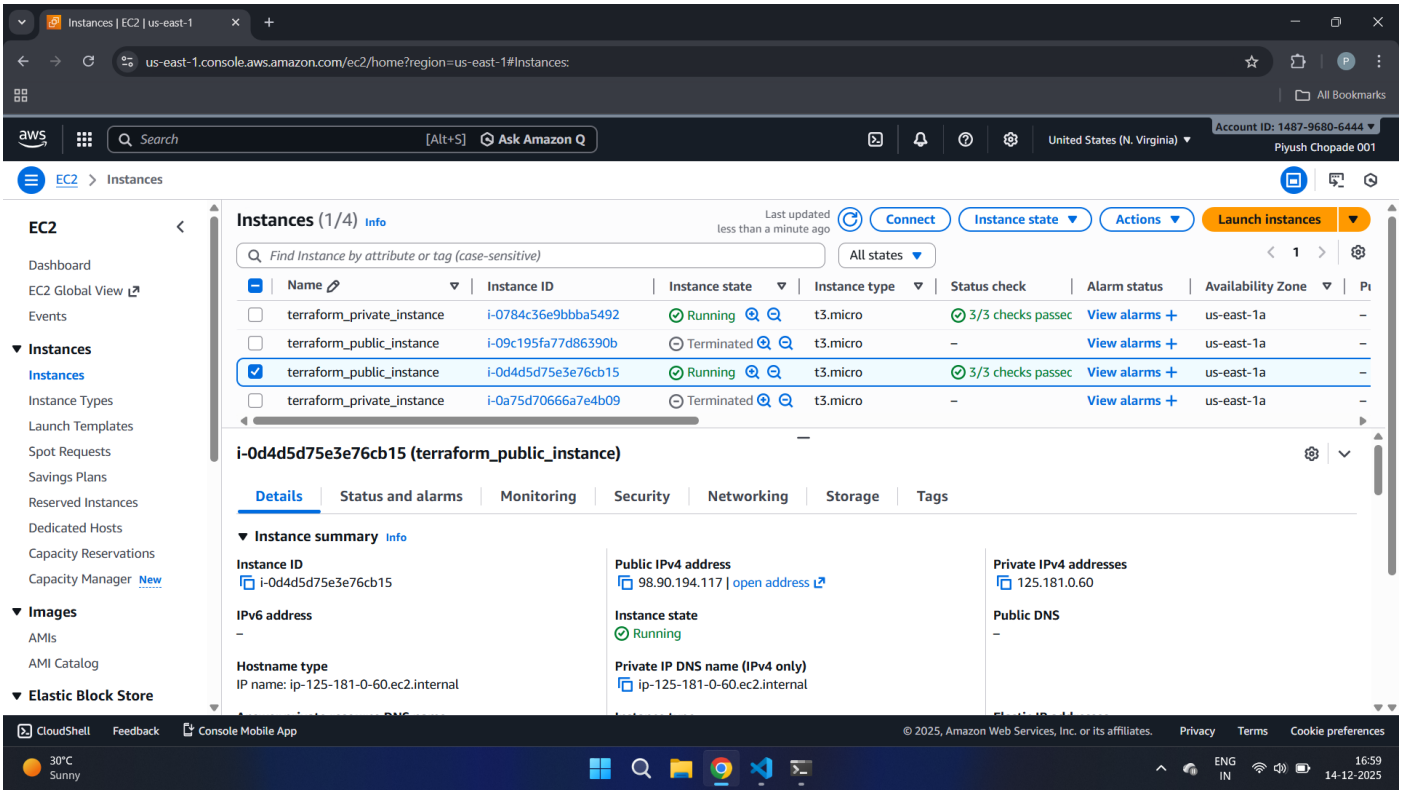
5. here we get access of private instance. then <ping google.com> for check internet.

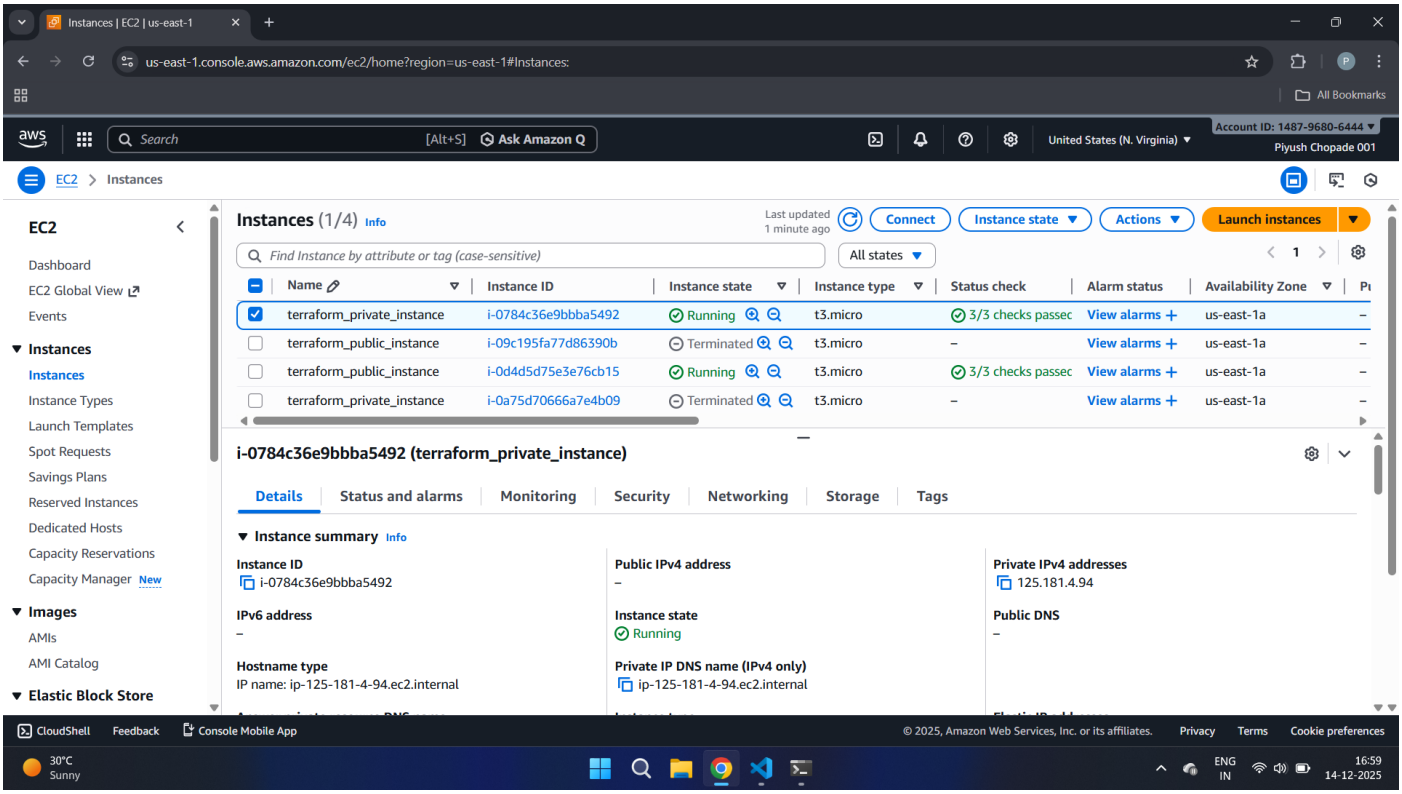
Project step-by-step-execution images.

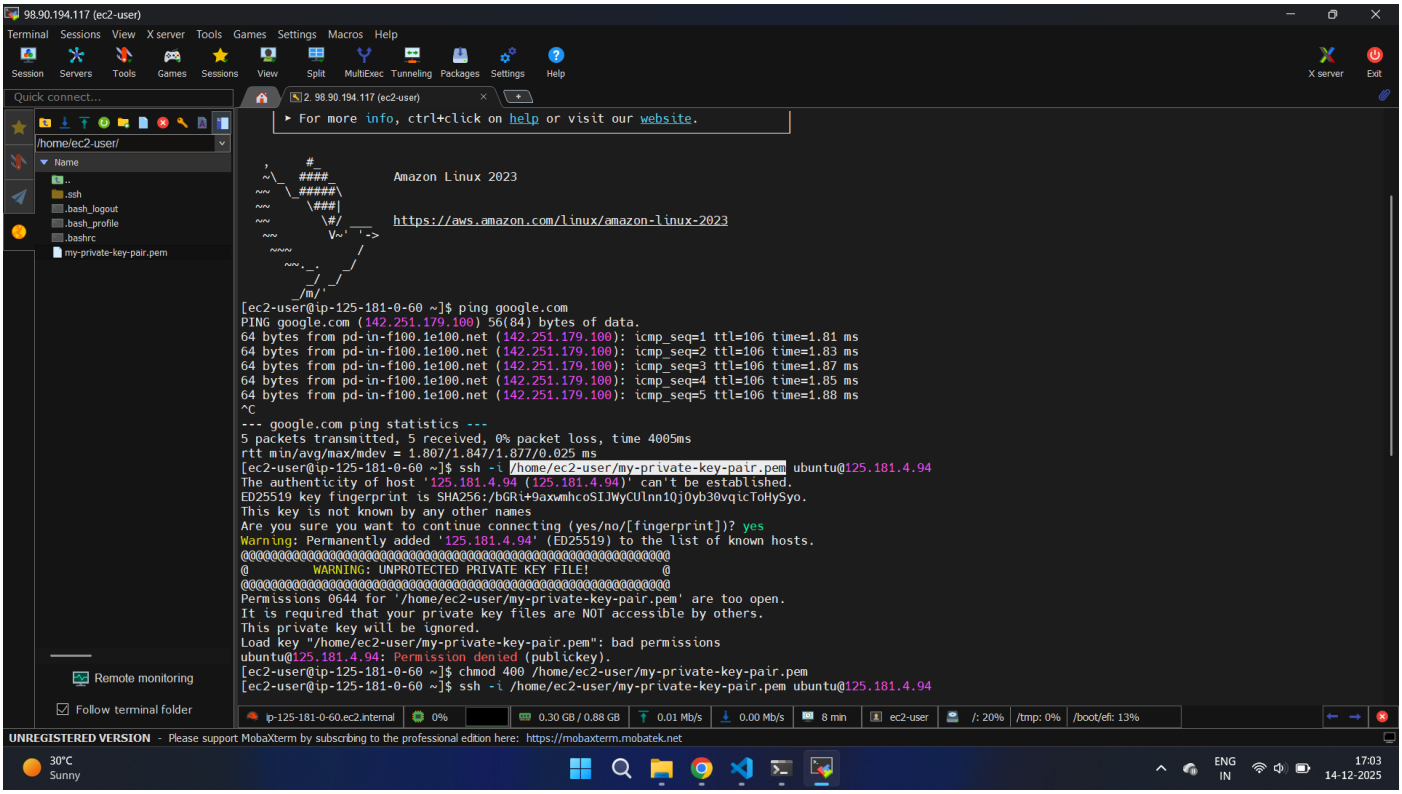


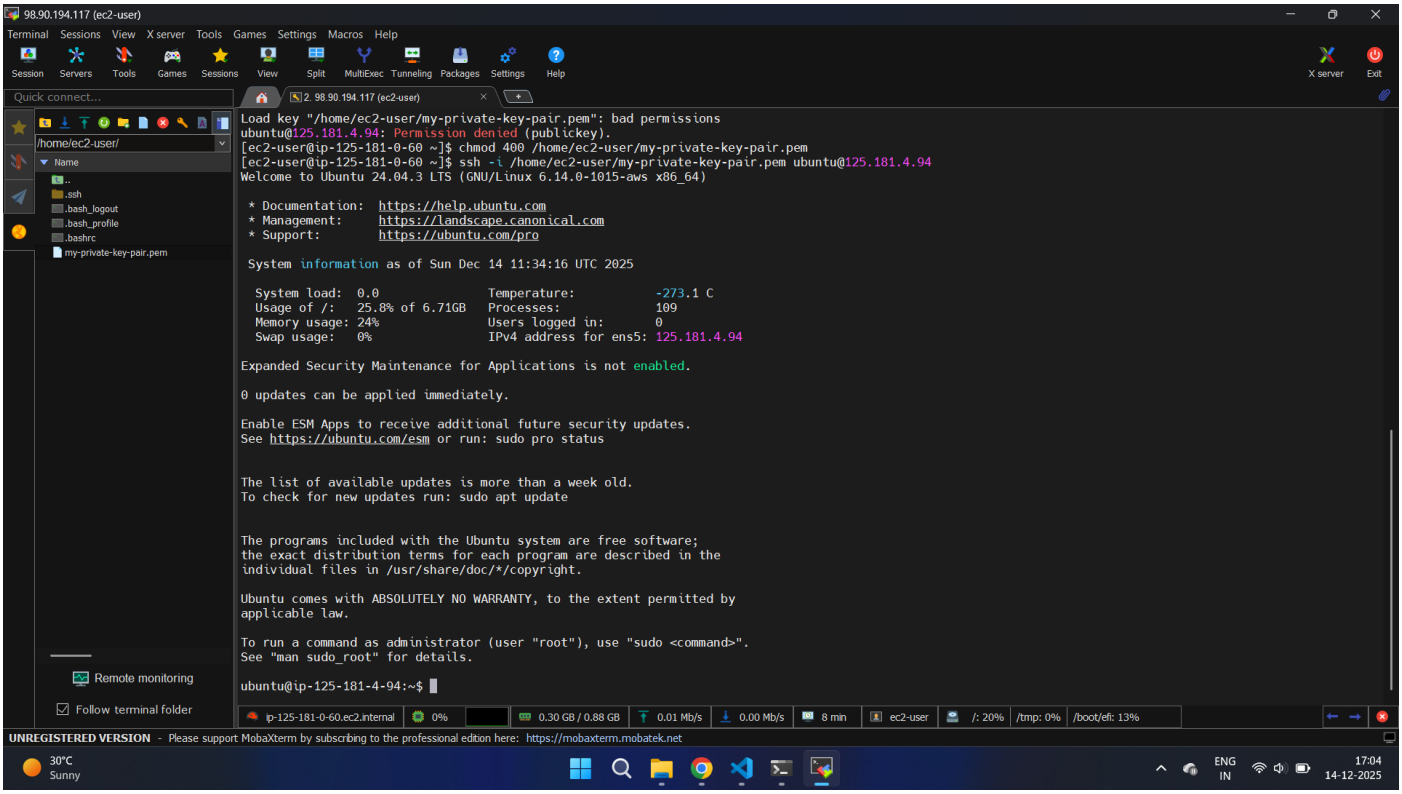

piyush_ubuntu@Piyush-Chopa
root@Piyush-Chopade: ~/terr
+ vpc_id = (known after apply)
}
aws_vpc.batch3 will be created
+ resource "aws_vpc" "batch3" {
+ arn = (known after apply)
+ cidr_block = "125.181.0.0/21"
+ default_network_acl_id = (known after apply)
+ default_route_table_id = (known after apply)
+ default_security_group_id = (known after apply)
+ dhcp_options_id = (known after apply)
+ enable_dns_hostnames = (known after apply)
+ enable_dns_support = true
+ enable_network_address_usage_metrics = (known after apply)
+ id = (known after apply)
+ instance_tenancy = "default"
+ ipv6_association_id = (known after apply)
+ ipv6_cidr_block = (known after apply)
+ ipv6_cidr_block_network_border_group = (known after apply)
+ main_route_table_id = (known after apply)
+ owner_id = (known after apply)
+ region = "us-east-1"
+ tags = {
+ "Name" = "batch2-tf-vpc-011125"
}
+ tags_all = {
+ "Name" = "batch2-tf-vpc-011125"
+ "practice" = "terraform"
}
}
Plan: 14 to add, 0 to change, 0 to destroy.
Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.
root@Piyush-Chopade:~/terraform# terraform apply --auto-approve
30°C
Sunny
ENG
IN
16:32
14-12-2025

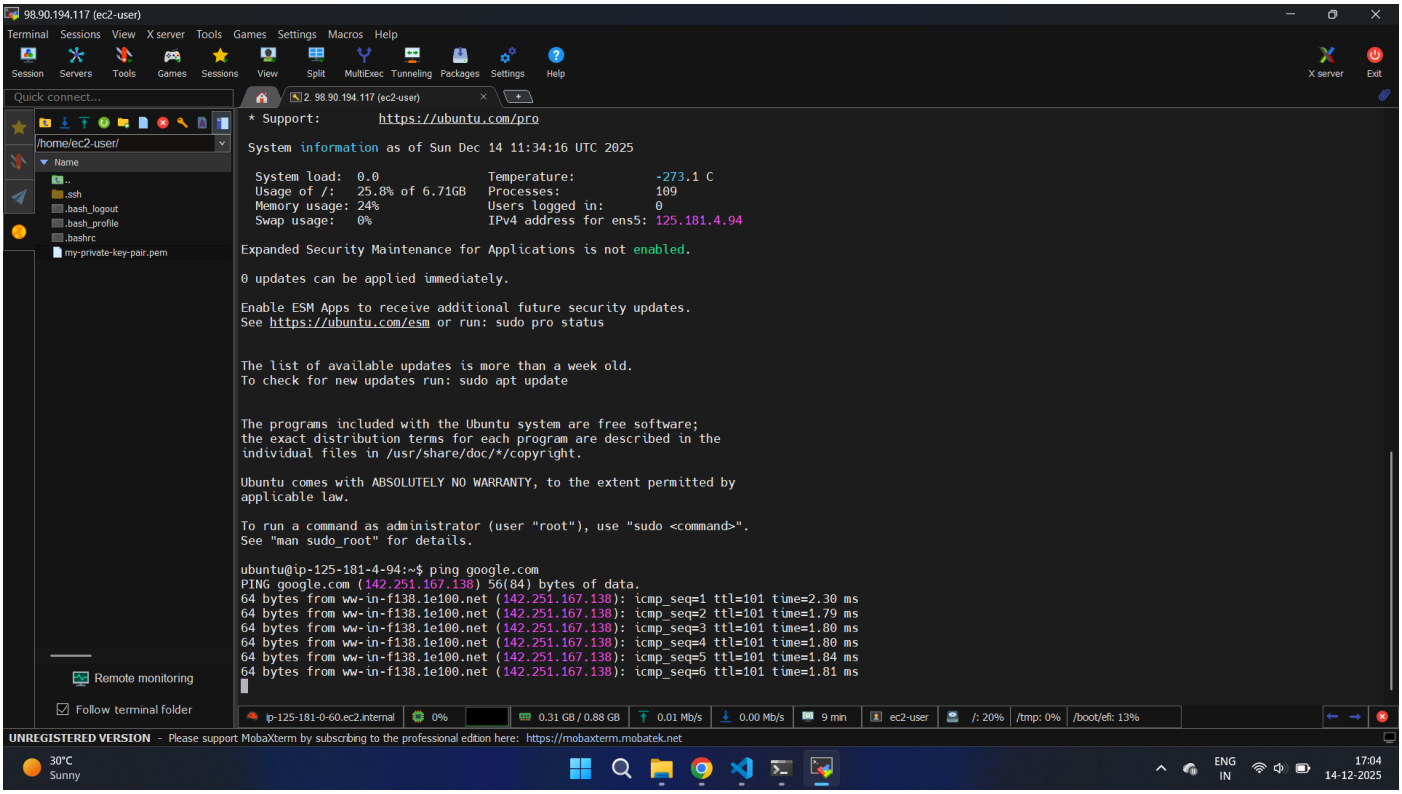

piyush_ubuntu@Piyush-Chopa
root@Piyush-Chopade: ~/terr
aws_vpc.batch3: Creation complete after 6s [id=vpc-066bf392956f21587]
aws_internet_gateway.igw: Creating...
aws_subnet.private: Creating...
aws_route_table.private_rt: Creating...
aws_security_group.piyush_ssh_sg: Creating...
aws_subnet.public: Creating...
aws_subnet.private: Creation complete after 2s [id=subnet-022b06a328289ce6c]
aws_subnet.public: Creation complete after 2s [id=subnet-09817c697be73ec4e]
aws_route_table_association.public_subnet_association: Creating...
aws_internet_gateway.igw: Creation complete after 2s [id=igw-0cebb1d2de70eb4fb]
aws_route.default_internet_route: Creating...
aws_nat_gateway.natgw: Creating...
aws_route_table.private_rt: Creation complete after 2s [id=rtb-020f9cfbd81831dea]
aws_route_table_association.private_subnet_association: Creating...
aws_route_table_association.public_subnet_association: Creation complete after 1s [id=rtbassoc-0791c3627bfe5410a]
aws_route_table_association.private_subnet_association: Creation complete after 2s [id=rtbassoc-0ed97086217003b44]
aws_route.default_internet_route: Creation complete after 5s [id=r-rtb-0dae5e654ea3224cf1080289494]
aws_security_group.piyush_ssh_sg: Creation complete after 9s [id=sg-0910ad8302fd65372]
aws_instance.private_instance: Creating...
aws_instance.public_instance: Creating...
aws_nat_gateway.natgw: Still creating... [00m13s elapsed]
aws_instance.public_instance: Still creating... [00m10s elapsed]
aws_instance.private_instance: Still creating... [00m10s elapsed]
aws_nat_gateway.natgw: Still creating... [00m23s elapsed]
aws_instance.private_instance: Creation complete after 17s [id=i-0784c36e9bbba5492]
aws_instance.public_instance: Creation complete after 17s [id=i-0d4d5d75e3e76cb15]
aws_nat_gateway.natgw: Still creating... [00m33s elapsed]
aws_nat_gateway.natgw: Still creating... [00m46s elapsed]
aws_nat_gateway.natgw: Still creating... [00m56s elapsed]
aws_nat_gateway.natgw: Still creating... [01m06s elapsed]
aws_nat_gateway.natgw: Still creating... [01m19s elapsed]
aws_nat_gateway.natgw: Still creating... [01m29s elapsed]
aws_nat_gateway.natgw: Creation complete after 1m39s [id=nat-0af8a176aebd1fc26]
aws_route.private_nat_route: Creating...
aws_route.private_nat_route: Creation complete after 2s [id=r-rtb-020f9cfbd81831dea1080289494]
Apply complete! Resources: 14 added, 0 changed, 0 destroyed.
root@Piyush-Chopade:~/terraform#
30°C
Sunny
ENG
IN
16:56
14-12-2025

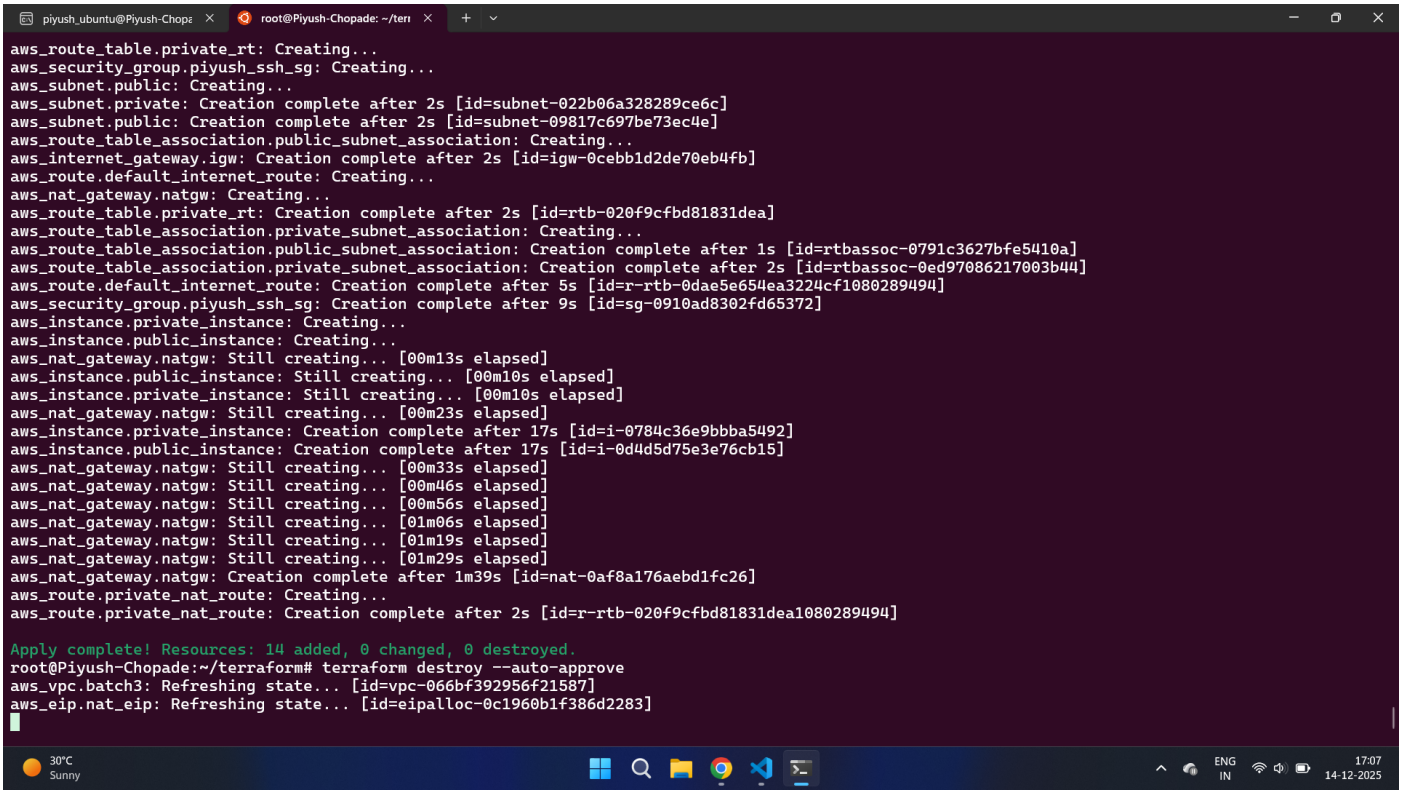

Instances | EC2 | us-east-1
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:
Search [Alt+S] Ask Amazon Q
United States (N. Virginia)
Account ID: 1487-9680-6444
Piyush Chopade 001
EC2 > Instances
EC2
Dashboard
EC2 Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Instances (1/4) Info
Last updated less than a minute ago
Connect
Instance state
Actions
Launch instances
Find Instance by attribute or tag (case-sensitive)
All states
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
terraform_private_instance i-0784c36e9bbba5492 Running t3.micro 3/3 checks passec View alarms us-east-1a
terraform_public_instance i-09c195fa77d86390b Terminated t3.micro – View alarms us-east-1a
terraform_public_instance i-0d4d5d75e3e76cb15 Running t3.micro 3/3 checks passec View alarms us-east-1a
terraform_private_instance i-0a75d70666a7e4b09 Terminated t3.micro – View alarms us-east-1a
i-0d4d5d75e3e76cb15 (terraform_public_instance)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary Info
Instance ID
i-0d4d5d75e3e76cb15
IPv6 address
–
Hostname type
IP name: ip-125-181-0-60.ec2.internal
Public IPv4 address
98.90.194.117 | open address
Instance state
Running
Private IP DNS name (IPv4 only)
ip-125-181-0-60.ec2.internal
Private IPv4 addresses
125.181.0.60
Public DNS
–
CloudShell
Feedback
Console Mobile App
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Privacy
Terms
Cookie preferences
30°C Sunny
ENG IN
16:59 14-12-2025

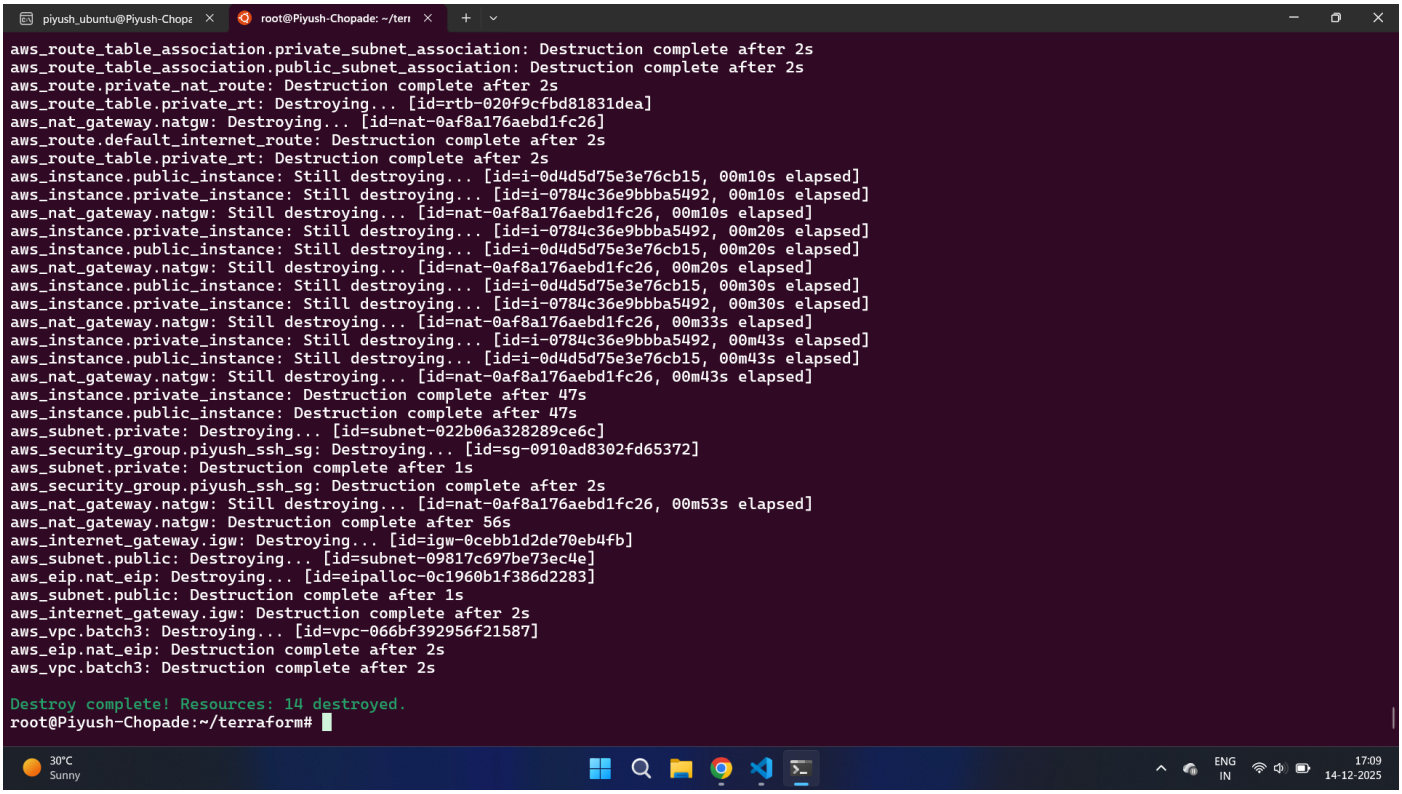

Instances | EC2 | us-east-1
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:
Search [Alt+S] Ask Amazon Q
United States (N. Virginia)
Account ID: 1487-9680-6444
Piyush Chopade 001
EC2 > Instances
EC2
Dashboard
EC2 Global View
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Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
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Instance type
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terraform_private_instance i-0a75d70666a7e4b09 Terminated t3.micro – View alarms us-east-1a
i-0784c36e9bbba5492 (terraform_private_instance)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary Info
Instance ID
i-0784c36e9bbba5492
IPv6 address
–
Hostname type
IP name: ip-125-181-4-94.ec2.internal
Public IPv4 address
–
Instance state
Running
Private IP DNS name (IPv4 only)
ip-125-181-4-94.ec2.internal
Private IPv4 addresses
125.181.4.94
Public DNS
–
CloudShell
Feedback
Console Mobile App
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Privacy
Terms
Cookie preferences
30°C Sunny
ENG IN
16:59 14-12-2025


98.90.194.117 (ec2-user)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
X server Exit
Quick connect...
2. 98.90.194.117 (ec2-user)
/home/ec2-user/
Name
.ssh
.bash_logout
.bash_profile
.bashrc
my-private-key-pair.pem
For more info, ctrl+click on help or visit our website.
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
[ec2-user@ip-125-181-0-60 ~]$ ping google.com
PING google.com (142.251.179.100) 56(84) bytes of data.
64 bytes from pd-in-f100.1e100.net (142.251.179.100): icmp_seq=1 ttl=106 time=1.81 ms
64 bytes from pd-in-f100.1e100.net (142.251.179.100): icmp_seq=2 ttl=106 time=1.83 ms
64 bytes from pd-in-f100.1e100.net (142.251.179.100): icmp_seq=3 ttl=106 time=1.87 ms
64 bytes from pd-in-f100.1e100.net (142.251.179.100): icmp_seq=4 ttl=106 time=1.85 ms
64 bytes from pd-in-f100.1e100.net (142.251.179.100): icmp_seq=5 ttl=106 time=1.88 ms
^C
--- google.com ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4005ms
rtt min/avg/max/mdev = 1.807/1.847/1.877/0.025 ms
[ec2-user@ip-125-181-0-60 ~]$ ssh -i /home/ec2-user/my-private-key-pair.pem ubuntu@125.181.4.94
The authenticity of host '125.181.4.94 (125.181.4.94)' can't be established.
ED25519 key fingerprint is SHA256:/bGRi+9axwmhcoSIJWyCUlnn1Qj0yb30vqicToHySyo.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '125.181.4.94' (ED25519) to the list of known hosts.
@ WARNING: UNPROTECTED PRIVATE KEY FILE! @
Permissions 0644 for '/home/ec2-user/my-private-key-pair.pem' are too open.
It is required that your private key files are NOT accessible by others.
This private key will be ignored.
Load key "/home/ec2-user/my-private-key-pair.pem": bad permissions
ubuntu@125.181.4.94: Permission denied (publickey).
[ec2-user@ip-125-181-0-60 ~]$ chmod 400 /home/ec2-user/my-private-key-pair.pem
[ec2-user@ip-125-181-0-60 ~]$ ssh -i /home/ec2-user/my-private-key-pair.pem ubuntu@125.181.4.94
Remote monitoring
Follow terminal folder
ip-125-181-0-60.ec2.internal 0% 0.30 GB / 0.88 GB 0.01 Mb/s 0.00 Mb/s 8 min ec2-user /: 20% /tmp: 0% /boot/efi: 13%
UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: https://mobaxterm.mobatek.net
30°C Sunny
ENG IN
17:03 14-12-2025


98.90.194.117 (ec2-user)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
X server Exit
Quick connect...
2. 98.90.194.117 (ec2-user)
/home/ec2-user/
Name
.ssh
.bash_logout
.bash_profile
.bashrc
my-private-key-pair.pem
Load key "/home/ec2-user/my-private-key-pair.pem": bad permissions
ubuntu@125.181.4.94: Permission denied (publickey).
[ec2-user@ip-125-181-0-60 ~]$ chmod 400 /home/ec2-user/my-private-key-pair.pem
[ec2-user@ip-125-181-0-60 ~]$ ssh -i /home/ec2-user/my-private-key-pair.pem ubuntu@125.181.4.94
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1015-aws x86_64)
* Documentation: https://help.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro
System information as of Sun Dec 14 11:34:16 UTC 2025
System load: 0.0 Temperature: -273.1 C
Usage of /: 25.8% of 6.71GB Processes: 109
Memory usage: 24% Users logged in: 0
Swap usage: 0% IPv4 address for ens5: 125.181.4.94
Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
The list of available updates is more than a week old.
To check for new updates run: sudo apt update
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
ubuntu@ip-125-181-4-94:~$
Remote monitoring
Follow terminal folder
ip-125-181-0-60.ec2.internal 0% 0.30 GB / 0.88 GB 0.01 Mb/s 0.00 Mb/s 8 min ec2-user /: 20% /tmp: 0% /boot/efi: 13%
UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: https://mobaxterm.mobatek.net
30°C Sunny
ENG IN
17:04 14-12-2025


98.90.194.117 (ec2-user)
Terminal Sessions View X server Tools Games Settings Macros Help
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The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
ubuntu@ip-125-181-4-94:~$ ping google.com
PING google.com (142.251.167.138) 56(84) bytes of data.
64 bytes from ww-in-f138.1e100.net (142.251.167.138): icmp_seq=1 ttl=101 time=2.30 ms
64 bytes from ww-in-f138.1e100.net (142.251.167.138): icmp_seq=2 ttl=101 time=1.79 ms
64 bytes from ww-in-f138.1e100.net (142.251.167.138): icmp_seq=3 ttl=101 time=1.80 ms
64 bytes from ww-in-f138.1e100.net (142.251.167.138): icmp_seq=4 ttl=101 time=1.80 ms
64 bytes from ww-in-f138.1e100.net (142.251.167.138): icmp_seq=5 ttl=101 time=1.84 ms
64 bytes from ww-in-f138.1e100.net (142.251.167.138): icmp_seq=6 ttl=101 time=1.81 ms
ip-125-181-0-60.ec2.internal 0% 0.31 GB / 0.88 GB 0.01 Mb/s 0.00 Mb/s 9 min ec2-user /: 20% /tmp: 0% /boot/efi: 13%
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30°C Sunny
ENG IN
17:04 14-12-2025


piyush_ubuntu@Piyush-Chopa
root@Piyush-Chopade: ~/terr
aws_route_table.private_rt: Creating...
aws_security_group.piyush_ssh_sg: Creating...
aws_subnet.public: Creating...
aws_subnet.private: Creation complete after 2s [id=subnet-022b06a328289ce6c]
aws_subnet.public: Creation complete after 2s [id=subnet-09817c697be73ec4e]
aws_route_table_association.public_subnet_association: Creating...
aws_internet_gateway.igw: Creation complete after 2s [id=igw-0cebb1d2de70eb4fb]
aws_route.default_internet_route: Creating...
aws_nat_gateway.natgw: Creating...
aws_route_table.private_rt: Creation complete after 2s [id=rtb-020f9cfbd81831dea]
aws_route_table_association.private_subnet_association: Creating...
aws_route_table_association.public_subnet_association: Creation complete after 1s [id=rtbassoc-0791c3627bfe5410a]
aws_route_table_association.private_subnet_association: Creation complete after 2s [id=rtbassoc-0ed97086217003b44]
aws_route.default_internet_route: Creation complete after 5s [id=r-rtb-0dae5e654ea3224cf1080289494]
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aws_instance.public_instance: Creating...
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aws_instance.public_instance: Still creating... [00m10s elapsed]
aws_instance.private_instance: Still creating... [00m10s elapsed]
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aws_nat_gateway.natgw: Still creating... [00m46s elapsed]
aws_nat_gateway.natgw: Still creating... [00m56s elapsed]
aws_nat_gateway.natgw: Still creating... [01m06s elapsed]
aws_nat_gateway.natgw: Still creating... [01m19s elapsed]
aws_nat_gateway.natgw: Still creating... [01m29s elapsed]
aws_nat_gateway.natgw: Creation complete after 1m39s [id=nat-0af8a176aebd1fc26]
aws_route.private_nat_route: Creating...
aws_route.private_nat_route: Creation complete after 2s [id=r-rtb-020f9cfbd81831dea1080289494]
Apply complete! Resources: 14 added, 0 changed, 0 destroyed.
root@Piyush-Chopade:~/terraform# terraform destroy --auto-approve
aws_vpc.batch3: Refreshing state... [id=vpc-066bf392956f21587]
aws_eip.nat_eip: Refreshing state... [id=eipalloc-0c1960b1f386d2283]
30°C Sunny
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17:07 14-12-2025


piyush_ubuntu@Piyush-Chopa
root@Piyush-Chopade: ~/terr
aws_route_table_association.private_subnet_association: Destruction complete after 2s
aws_route_table_association.public_subnet_association: Destruction complete after 2s
aws_route.private_nat_route: Destruction complete after 2s
aws_route_table.private_rt: Destroying... [id=rtb-020f9cfbd81831dea]
aws_nat_gateway.natgw: Destroying... [id=nat-0af8a176aebd1fc26]
aws_route.default_internet_route: Destruction complete after 2s
aws_route_table.private_rt: Destruction complete after 2s
aws_instance.public_instance: Still destroying... [id=i-0d4d5d75e3e76cb15, 00m10s elapsed]
aws_instance.private_instance: Still destroying... [id=i-0784c36e9bbba5492, 00m10s elapsed]
aws_nat_gateway.natgw: Still destroying... [id=nat-0af8a176aebd1fc26, 00m10s elapsed]
aws_instance.private_instance: Still destroying... [id=i-0784c36e9bbba5492, 00m20s elapsed]
aws_instance.public_instance: Still destroying... [id=i-0d4d5d75e3e76cb15, 00m20s elapsed]
aws_nat_gateway.natgw: Still destroying... [id=nat-0af8a176aebd1fc26, 00m20s elapsed]
aws_instance.public_instance: Still destroying... [id=i-0d4d5d75e3e76cb15, 00m30s elapsed]
aws_instance.private_instance: Still destroying... [id=i-0784c36e9bbba5492, 00m30s elapsed]
aws_nat_gateway.natgw: Still destroying... [id=nat-0af8a176aebd1fc26, 00m33s elapsed]
aws_instance.private_instance: Still destroying... [id=i-0784c36e9bbba5492, 00m43s elapsed]
aws_instance.public_instance: Still destroying... [id=i-0d4d5d75e3e76cb15, 00m43s elapsed]
aws_nat_gateway.natgw: Still destroying... [id=nat-0af8a176aebd1fc26, 00m43s elapsed]
aws_instance.private_instance: Destruction complete after 47s
aws_instance.public_instance: Destruction complete after 47s
aws_subnet.private: Destroying... [id=subnet-022b06a328289ce6c]
aws_security_group.piyush_ssh_sg: Destroying... [id=sg-0910ad8302fd65372]
aws_subnet.private: Destruction complete after 1s
aws_security_group.piyush_ssh_sg: Destruction complete after 2s
aws_nat_gateway.natgw: Still destroying... [id=nat-0af8a176aebd1fc26, 00m53s elapsed]
aws_nat_gateway.natgw: Destruction complete after 56s
aws_internet_gateway.igw: Destroying... [id=igw-0cebb1d2de70eb4fb]
aws_subnet.public: Destroying... [id=subnet-09817c697be73ec4e]
aws_eip.nat_eip: Destroying... [id=eipalloc-0c1960b1f386d2283]
aws_subnet.public: Destruction complete after 1s
aws_internet_gateway.igw: Destruction complete after 2s
aws_vpc.batch3: Destroying... [id=vpc-066bf392956f21587]
aws_eip.nat_eip: Destruction complete after 2s
aws_vpc.batch3: Destruction complete after 2s
Destroy complete! Resources: 14 destroyed.
root@Piyush-Chopade:~/terraform#
30°C
Sunny
ENG
IN
17:09
14-12-2025